

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\frac{d\rho}{dt} = \frac{1}{i\hbar} \left(\begin{pmatrix} 0 & 1 \\ q & 0 \end{pmatrix} \rho \begin{pmatrix} q^{\frac{1}{2}} & 0 \\ 0 & q^{-\frac{1}{2}} \end{pmatrix} - \begin{pmatrix} q^{\frac{1}{2}} & 0 \\ 0 & q^{-\frac{1}{2}} \end{pmatrix} \rho \begin{pmatrix} 0 & q \\ 1 & 0 \end{pmatrix} \right),$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} (\Delta v)_{\sigma,\tau} &= (v_{\sigma})[(c_{\sigma})(^{\sigma}v_{\tau})(c_{\sigma})^{-1}](v_{\sigma\tau})^{-1} \\ &= (v_{\sigma})(c_{\sigma})(^{\sigma}v_{\tau})[(^{\sigma}c_{\tau})(c_{\sigma\tau})^{-1}](v_{\sigma\tau})^{-1} \\ &= (b_{\sigma})(^{\sigma}b_{\tau})(b_{\sigma\tau})^{-1} = (\Delta b)_{\sigma,\tau} \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &\lfloor \frac{-2mp^i}{q-1} \rfloor + \lfloor \frac{mdp^i}{q-1} \rfloor + \lfloor \frac{-m(d-1)p^i}{q-1} \rfloor - \lfloor \frac{-mp^i}{q-1} \rfloor + 1 \\ &= \sum_{h=1}^{d-2} \lfloor \langle \frac{hp^i}{d-1} \rangle - \frac{mp^i}{q-1} \rfloor + \sum_{\substack{h=1 \\ h \neq \frac{d}{2}}}^{d-1} \lfloor \langle \frac{-hp^i}{d} \rangle + \frac{mp^i}{q-1} \rfloor. \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &\frac{d}{dt} [V(x(t)) + Q_k(t - t_k, x_k)] + 2\alpha V(x(t)) + 2\alpha Q_k(t - t_k, x_k) \\ &= \nabla V(x)^T f(x(t), x(t_k)) + \nabla_x F_k(t - t_k, x(t), x(t_{k+1}), x(t_k))^T f(x) \\ &+ \frac{d}{dt} F_k(t - t_k, x(t), x(t_{k+1}), x(t_k)) + 2\alpha V(x(t)) \\ &+ 2\alpha F_k(t - t_k, x(t), x(t_{k+1}), x(t_k)) < -\delta \|x(t)\| \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
\|r_h^+ u - r_h u\|_{0,2} &\leq Ch^{\sigma'} \left[h\|r_h^+ u - u\|_{1,\eta} + h\|r_h u - u\|_{1,\eta} + \|r_h^+ u - u\|_{0,\eta} + \|r_h u - u\|_{0,\eta} \right. \\
&\quad \left. + (h^\mu \|r_h u - u\|_{\mu,\eta})^{1/2} (\|r_h^+ u - u\|_{0,\eta} + \|r_h u - u\|_{0,\eta})^{1/2} \right. \\
&\quad \left. + h^\mu \|r_h u - u\|_{\mu,\eta} \right],
\end{aligned}$$